

TEMPERATE CONTINENTAL STEPPE CLIMATE

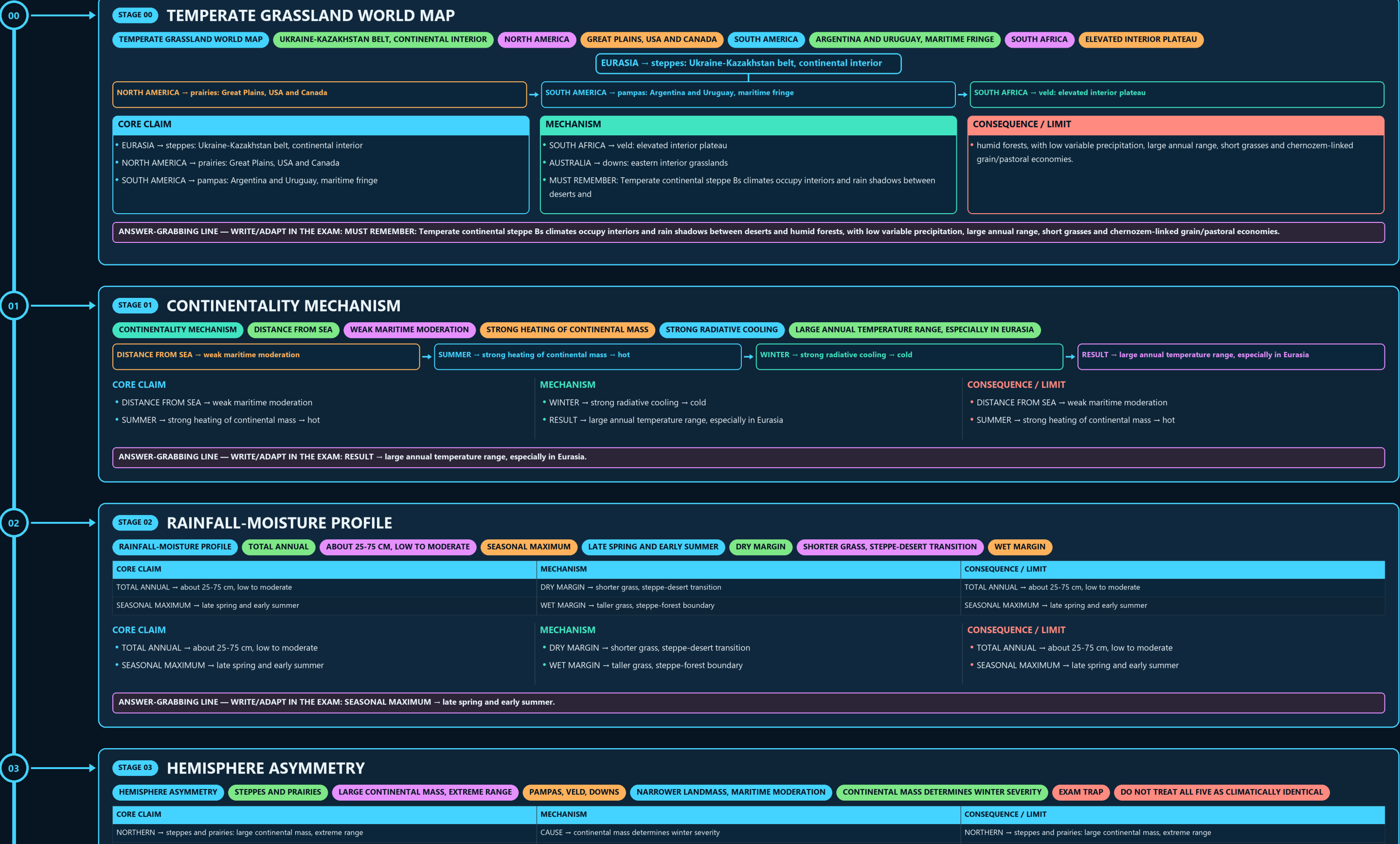
TEMPERATE GRASSLAND WORLD MAP → CONTINENTALITY MECHANISM → RAINFALL-MOISTURE PROFILE → HEMISPHERE ASYMMETRY → CHERNOZEM-SOIL CHAIN → GRANARY CONVERSION TIMELINE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



02

STAGE 02

RAINFALL-MOISTURE PROFILE

RAINFALL-MOISTURE PROFILETOTAL ANNUALABOUT 25-75 CM, LOW TO MODERATESEASONAL MAXIMUMLATE SPRING AND EARLY SUMMERDRY MARGINSHORTER GRASS, STEPPE-DESERT TRANSITIONWET MARGIN

CORE CLAIM

TOTAL ANNUAL → about 25-75 cm, low to moderate

SEASONAL MAXIMUM → late spring and early summer

MECHANISM

DRY MARGIN → shorter grass, steppe-desert transition

WET MARGIN → taller grass, steppe-forest boundary

CONSEQUENCE / LIMIT

TOTAL ANNUAL → about 25-75 cm, low to moderate

SEASONAL MAXIMUM → late spring and early summer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEASONAL MAXIMUM → late spring and early summer.

03

STAGE 03

HEMISPHERE ASYMMETRY

HEMISPHERE ASYMMETRYSTEPPESS AND PRAIRIESLARGE CONTINENTAL MASS, EXTREME RANGE

PAMPAS, VELD, DOWNSNARROWER LANDMASS, MARITIME MODERATIONCONTINENTAL MASS DETERMINES WINTER SEVERITYEXAM TRAPDO NOT TREAT ALL FIVE AS CLIMATICALLY IDENTICAL

CORE CLAIM

NORTHERN → steppes and prairies: large continental mass, extreme range

MECHANISM

CAUSE → continental mass determines winter severity

CONSEQUENCE / LIMIT

NORTHERN → steppes and prairies: large continental mass, extreme range

03

STAGE 03

HEMISPHERE ASYMMETRY

HEMISPHERE ASYMMETRY

STEPPE AND PRAIRIES

LARGE CONTINENTAL MASS, EXTREME RANGE

PAMPAS, VELD, DOWNS

NARROWER LANDMASS, MARITIME MODERATION

CONTINENTAL MASS DETERMINES WINTER SEVERITY

EXAM TRAP

DO NOT TREAT ALL FIVE AS CLIMATICALLY IDENTICAL

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NORTHERN → steppes and prairies: large continental mass, extreme range	CAUSE → continental mass determines winter severity	NORTHERN → steppes and prairies: large continental mass, extreme range
SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation	EXAM TRAP → do not treat all five as climatically identical	SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

CORE CLAIM

• NORTHERN → steppes and prairies: large continental mass, extreme range

• SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

MECHANISM

• CAUSE → continental mass determines winter severity

• EXAM TRAP → do not treat all five as climatically identical

CONSEQUENCE / LIMIT

• NORTHERN → steppes and prairies: large continental mass, extreme range

• SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEASONAL MAXIMUM → late spring and early summer.

04

STAGE 04

CHERNOZEM-SOIL CHAIN

CHERNOZEM-SOIL CHAIN

GRASSLAND COVER

DEEP ROOT NETWORK DECAYS IN PLACE

LOW LEACHING

BASE-RICH CONDITIONS PRESERVED

HUMUS ACCUMULATES

DARK, DEEP, FERTILE SOIL PROFILE

CHERNOZEM OR BLACK EARTH, AMONG WORLD'S MOST FERTILE

GRASSLAND COVER → deep root network decays in place

LOW LEACHING → base-rich conditions preserved

HUMUS ACCUMULATES → dark, deep, fertile soil profile

RESULT → chernozem or black earth, among world's most fertile

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
GRASSLAND COVER → deep root network decays in place	HUMUS ACCUMULATES → dark, deep, fertile soil profile	GRASSLAND COVER → deep root network decays in place
LOW LEACHING → base-rich conditions preserved	RESULT → chernozem or black earth, among world's most fertile	LOW LEACHING → base-rich conditions preserved

CORE CLAIM

• GRASSLAND COVER → deep root network decays in place

• LOW LEACHING → base-rich conditions preserved

MECHANISM

• HUMUS ACCUMULATES → dark, deep, fertile soil profile

• RESULT → chernozem or black earth, among world's most fertile

CONSEQUENCE / LIMIT

• GRASSLAND COVER → deep root network decays in place

• LOW LEACHING → base-rich conditions preserved

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → chernozem or black earth, among world's most fertile.

05

STAGE 05

GRANARY CONVERSION TIMELINE

GRANARY CONVERSION TIMELINE

NATURAL STATE

PASTORAL GRASSLAND, LOW-DENSITY USE

RAIL + OCEAN FREIGHT

ACCESS TO DISTANT INDUSTRIAL MARKETS

MECHANISATION + HYV

HIGH OUTPUT PER WORKER ON LEVEL PLAINS

WORLD'S WHEAT GRANARIES

NATURAL STATE → pastoral grassland, low-density use

RAIL + OCEAN FREIGHT → access to distant industrial markets

MECHANISATION + HYV → high output per worker on level plains

RESULT → world's wheat granaries: Ukraine, prairies, pampas

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
NATURAL STATE → pastoral grassland, low-density use	MECHANISATION + HYV → high output per worker on level plains	NATURAL STATE → pastoral grassland, low-density use
RAIL + OCEAN FREIGHT → access to distant industrial markets	RESULT → world's wheat granaries: Ukraine, prairies, pampas	RAIL + OCEAN FREIGHT → access to distant industrial markets

CORE CLAIM

• NATURAL STATE → pastoral grassland, low-density use

• RAIL + OCEAN FREIGHT → access to distant industrial markets

MECHANISM

• MECHANISATION + HYV → high output per worker on level plains

• RESULT → world's wheat granaries: Ukraine, prairies, pampas

CONSEQUENCE / LIMIT

• NATURAL STATE → pastoral grassland, low-density use

• RAIL + OCEAN FREIGHT → access to distant industrial markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → world's wheat granaries: Ukraine, prairies, pampas.

06

STAGE 06

ANTI-DETERMINISM FIREWALL

ANTI-DETERMINISM FIREWALL

PHYSICAL BASE

FERTILE SOIL + LEVEL RELIEF + SUITABLE CLIMATE

CONVERSION AGENTS

RAIL, FREIGHT, MECHANISATION, MARKET DEMAND

WITHOUT ACCESS

SAME SOIL, NO GRANARY FOR CENTURIES

ENDOWMENT IS NECESSARY, ACCESS IS SUFFICIENT

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand

WITHOUT ACCESS → same soil, no granary for centuries

CONCLUSION → endowment is necessary, access is sufficient

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
PHYSICAL BASE → fertile soil + level relief + suitable climate	WITHOUT ACCESS → same soil, no granary for centuries	PHYSICAL BASE → fertile soil + level relief + suitable climate
CONVERSION AGENTS → rail, freight, mechanisation, market demand	CONCLUSION → endowment is necessary, access is sufficient	CONVERSION AGENTS → rail, freight, mechanisation, market demand

CORE CLAIM

• PHYSICAL BASE → fertile soil + level relief + suitable climate

• CONVERSION AGENTS → rail, freight, mechanisation, market demand

MECHANISM

• WITHOUT ACCESS → same soil, no granary for centuries

• CONCLUSION → endowment is necessary, access is sufficient

CONSEQUENCE / LIMIT

• PHYSICAL BASE → fertile soil + level relief + suitable climate

• CONVERSION AGENTS → rail, freight, mechanisation, market demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → world's wheat granaries: Ukraine, prairies, pampas.

06

STAGE 06 ANTI-DETERMINISM FIREWALL

ANTI-DETERMINISM FIREWALL

PHYSICAL BASE

FERTILE SOIL + LEVEL RELIEF + SUITABLE CLIMATE

CONVERSION AGENTS

RAIL, FREIGHT, MECHANISATION, MARKET DEMAND

WITHOUT ACCESS

SAME SOIL, NO GRANARY FOR CENTURIES

ENDOWMENT IS NECESSARY, ACCESS IS SUFFICIENT

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand

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CONCLUSION → endowment is necessary, access is sufficient

CORE CLAIM

- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

MECHANISM

- WITHOUT ACCESS → same soil, no granary for centuries
- CONCLUSION → endowment is necessary, access is sufficient

CONSEQUENCE / LIMIT

- PHYSICAL BASE → fertile soil + level relief + suitable climate
- CONVERSION AGENTS → rail, freight, mechanisation, market demand

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CONCLUSION → endowment is necessary, access is sufficient.

07

STAGE 07 ECOLOGICAL-COST CHAIN

ECOLOGICAL-COST CHAIN

SOD REMOVAL

BARE FINE SOIL EXPOSED TO WIND

DROUGHT YEAR

VEGETATION FAILS, DEFLATION PEAKS

ORGANIC DECLINE

REDUCED HUMUS, COMPACTION RISK

AQUIFER DRAWDOWN

SOD REMOVAL → bare fine soil exposed to wind

DROUGHT YEAR → vegetation fails, deflation peaks

ORGANIC DECLINE → reduced humus, compaction risk

AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

CORE CLAIM

- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

MECHANISM

- ORGANIC DECLINE → reduced humus, compaction risk
- AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater

CONSEQUENCE / LIMIT

- SOD REMOVAL → bare fine soil exposed to wind
- DROUGHT YEAR → vegetation fails, deflation peaks

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: AQUIFER DRAWDOWN → irrigation in semi-arid belt depletes groundwater.

08

STAGE 08 INDIA WHEAT GRANARY MAP

INDIA WHEAT GRANARY MAP

PUNJAB-HARYANA-WESTERN UP

INDO-GANGETIC ALLUVIAL PLAIN

ALLUVIAL, NOT CHERNOZEM (FUNDAMENTAL DIFFERENCE)

SOWN OCT-DEC, HARVESTED MAR-APR

HYV SEEDS + IRRIGATION + FERTILISER + MSP PROCUREMENT

PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain

SOIL → alluvial, not chernozem (fundamental difference)

SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr

ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

CORE CLAIM

- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

MECHANISM

- SEASON → Rabi: sown Oct-Dec, harvested Mar-Apr
- ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement

CONSEQUENCE / LIMIT

- PUNJAB-HARYANA-WESTERN UP → Indo-Gangetic alluvial plain
- SOIL → alluvial, not chernozem (fundamental difference)

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement.

09

STAGE 09 GREEN REVOLUTION PACKAGE

GREEN REVOLUTION PACKAGE

HYV SEMI-DWARF WHEAT

RAISED YIELD POTENTIAL

CANAL + TUBE-WELL IRRIGATION

REDUCED RAINFALL DEPENDENCE

CHEMICAL FERTILISERS

SUPPORTED INTENSIVE CULTIVATION

ASSURED FLOOR PRICE AND CENTRAL POOL STOCKS

HYV SEMI-DWARF WHEAT → raised yield potential

CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

CHEMICAL FERTILISERS → supported intensive cultivation

MSP/PROCUREMENT → assured floor price and central pool stocks

CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across

CORE CLAIM

- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

MECHANISM

- CHEMICAL FERTILISERS → supported intensive cultivation
- MSP/PROCUREMENT → assured floor price and central pool stocks

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all
- steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

09

STAGE 09 GREEN REVOLUTION PACKAGE

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HYV SEMI-DWARF WHEAT

RAISED YIELD POTENTIAL

CANAL + TUBE-WELL IRRIGATION

REDUCED RAINFALL DEPENDENCE

CHEMICAL FERTILISERS

SUPPORTED INTENSIVE CULTIVATION

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CORE CLAIM

- HYV SEMI-DWARF WHEAT → raised yield potential
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MECHANISM

- CHEMICAL FERTILISERS → supported intensive cultivation
- MSP/PROCUREMENT → assured floor price and central pool stocks

CONSEQUENCE / LIMIT

- CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Steppe differs from tropical savanna and humid prairie, chernozem is not universal across all steppes, and continentality concerns land-dominated thermal range rather than simply distance from a national coastline.

10

STAGE 10 SAVANNA-STEPPE COMPARISON

SAVANNA-STEPPE COMPARISON

TROPICAL SAVANNA VERSUS TEMPERATE STEPPE/PRAIRIE

WET-AND-DRY SEASONAL VERSUS LOW-TOTAL SPRING MAXIMUM

LEACHED LATERISING VERSUS BASE-RICH CHERNOZEM/PRAIRIE

PASTORAL-SUBSISTENCE VERSUS MECHANISED-COMMERCIAL GRAIN

EVIDENCE LIMIT

GRAIN PRODUCTIVITY DEPENDS ON SOIL, SNOWMELT, TECHNOLOGY AND MARKETS

CORE CLAIM

LATITUDE → tropical savanna versus temperate steppe/prairie

RAINFALL → wet-and-dry seasonal versus low-total spring maximum

MECHANISM

SOIL → leached laterising versus base-rich chernozem/prairie

ECONOMY → pastoral-subsistence versus mechanised-commercial grain

CONSEQUENCE / LIMIT

EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate

India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

CORE CLAIM

- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

MECHANISM

- SOIL → leached laterising versus base-rich chernozem/prairie
- ECONOMY → pastoral-subsistence versus mechanised-commercial grain

CONSEQUENCE / LIMIT

- EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate
- India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Grain productivity depends on soil, snowmelt, technology and markets as well as climate; India's wheat belt is a functional comparison with monsoonal irrigation and subtropical seasonality, not a climatic identity.

11

STAGE 11 STEPPE ANSWER SPINE

STEPPE ANSWER SPINE

CONTINENTAL INTERIOR GRASSLAND WITH LARGE ANNUAL RANGE

FIVE REGIONAL NAMES ON A WORLD MAP

MOISTURE DEFICIT, CHERNOZEM, GRANARY CONVERSION

ACCESS CONVERTED ENDOWMENT INTO RESOURCE

DEFINE → continental interior grassland with large annual range

LOCATE → five regional names on a world map

EXPLAIN → moisture deficit, chernozem, granary conversion

QUALIFY → anti-determinism: access converted endowment into resource

CORE CLAIM

- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

MECHANISM

- EXPLAIN → moisture deficit, chernozem, granary conversion
- QUALIFY → anti-determinism: access converted endowment into resource

CONSEQUENCE / LIMIT

- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → anti-determinism: access converted endowment into resource.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S WHEAT GRANARY

PUNJAB, HARYANA, WESTERN UP

INDO-GANGETIC PLAIN.

PUNJAB IS DESCRIBED AS A MAJOR FOOD BASKET/GRANARY IN

PUNJAB'S HIGH CULTIVATED SHARE AND CROPPING INTENSITY ARE SOURCE-ERA

HARYANA'S WHEAT BELT SPANS IRRIGATED EASTERN, CENTRAL AND WESTERN

WHEAT IS A RABI CROP

OPTIONAL SOURCE DEPTH

- India's wheat granary = Punjab, Haryana, western UP / Indo-Gangetic plain.
- Punjab is described as a major food basket/granary in the source text.
- Punjab's high cultivated share and cropping intensity are source-era indicators, not current statistics.
- Haryana's wheat belt spans irrigated eastern, central and western plains; use current district maps.

ADVANCED DEBATE

- Wheat is a Rabi crop: sown Oct–Dec, harvested Mar–Apr in source notes.
- Green Revolution package: HYV + irrigation + fertiliser + MSP/procurement.
- Soil is alluvial, not steppe chernozem.
- The source-era text records a substantial marketed surplus; current share requires a dated source.

LEGACY / NUANCE

- Surplus states: Punjab, Haryana, UP, Rajasthan, MP.
- Deficit destinations include Maharashtra, West Bengal, Bihar and Delhi.
- NOT: India's wheat belt sits on chernozem like the steppe → It sits on Indo-Gangetic alluvial soil.
- NOT: Wheat is a Kharif monsoon crop → Wheat is a Rabi winter crop.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.